EXERCISE 11.3

Integrate the following with respect to x:

(1) $(x+4)^5+\frac{5}{(2-5x)^4}-\operatorname{cosec}^2(3x-1)$

(2) $4\cos(5-2x)+9e^{3x-6}+\frac{24}{6-4x}$

(3) $\sec^2\frac{x}{5}+18\cos 2x+10\sec(5x+3)\tan(5x+3)$

(4) $\frac{8}{\sqrt{1-(4x)^2}}+\frac{27}{\sqrt{1-9x^2}}-\frac{15}{1+25x^2}$

(5) $\frac{6}{1+(3x+2)^2}-\frac{12}{\sqrt{1-(3-4x)^2}}$

(6) $\frac{1}{3}\cos\left(\frac{x}{3}-4\right)+\frac{7}{7x+9}+e^{\frac{x}{5}+3}$

11.6 Simple applications

So far in this section we have been using x as the variable of integration. In the case of applications, it is often convenient to use a different variable. For instance in the equation of motion the independent variable is time and the variable of integration is t.

In this section we discuss how integration is used to find the position and velocity of an object, given its acceleration and similar types of problems. Mathematically, this means that, starting with the derivative of a function, we must find the original function. Many common word which indicate derivative such as rate, growth, decay, marginal, change, varies, increase, decrease etc.

Example 11.10

If $f'(x)=3x^2-4x+5$ and $f(1)=3$, then find $f(x)$.

Solution

Given that $f'(x)=\frac{d}{dx}(f(x))=3x^2-4x+5$

Integrating on both sides with respect to x, we get

$$\int f'(x)\,dx=\int(3x^2-4x+5)\,dx$$
$$f(x)=x^3-2x^2+5x+c$$

To determine the constant of integration c, we have to apply the given information $f(1)=3$

$$f(1)=3\Rightarrow\ 3=(1)^3-2(1)^2+5(1)+c\Rightarrow c=-1$$

Thus $f(x)=x^3-2x^2+5x-1$.

Example 11.11

A train started from Madurai Junction towards Coimbatore at 3pm (time $t=0$) with velocity $v(t)=20t+50$ kilometre per hour, where t is measured in hours. Find the distance covered by the train at 5pm.

Solution

In calculus terminology, velocity $v=\frac{ds}{dt}$ is rate of change of position with time, where s is the distance.The velocity of the train is given by

$$v(t)=20t+50$$

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